Enzo-Fabrizio Calcagno-Miranda

LinkedIn | enzo_calcagno@hotmail.com | +1 (514) 713 7452 | Montreal, Quebec, Canada

EDUCATION

McGill University · B.Sc. in Computer Science (GPA: 3.72/4.0) Sep. 2022 - May. 2024

McGill University · B.Eng. in Electrical Engineering (transferred) Sep. 2018 - May. 2022

SKILLS

- **Languages:** Python, Java, Spring Boot, HTML, CSS, Javascript, C
- **Tech:** ReactJS, NodeJS, Git, PostgreSQL, Pandas, NumPy, Pydantic

WORK EXPERIENCE

Amazon May. 2023 - Aug. 2023

Full Stack Developer Intern

- Developed a **React/Java** tax dictionary tool for **200+ users** with field-level access control for precise permission visibility.
- Moved dynamic config to **AWS** AppConfig and offloaded heavy transforms to **AWS** Lambda, reducing average endpoint latency and enabling safe feature flags/quick rollback.
- Integrated **React Idle Timer** to emit usage metrics to a dashboard used by 30+ teammates.

Ericsson Jan. 2022 - Aug. 2022

Software Developer Intern

- Implemented **7+** features in **AngularJS** for Mimer, Ericsson's product lifecycle management tool, supporting **100+ team members**.
- Developed **50+ automated QA tests** in **Java** using **Selenium** to improve software reliability.
- Worked within **Agile** frameworks to ensure iterative development and continuous delivery.

PROJECTS

Music Academy Feedback Platform October 2025

Personal/Business Project

- Built a service + web UI in **Python** (**FastAPI**) and **React** for an online guitar academy: ingests lesson videos with **FFmpeg**, transcribes with **Whisper**, generates structured feedback using an **LLM**, supports instructor review/edit, and posts to **Discord**.
- Implemented a **GraphRAG** history aware layer using **Postgres** that retrieves each student's prior submissions to surface recurring patterns, highlight improvements, and suggest the next practice step in the review UI.

Collaborative Document Editor February 2025

Personal Project

- Built a real-time collaborative editor supporting concurrent users with **WebSocket** connections and **Protobuf** serialization for efficient binary message passing between browser and **Node.js** server.
- Implemented end-to-end encryption where clients maintain document state locally using operational transformation, with server acting only as encrypted message relay, ensuring zero-knowledge architecture.
- Developed responsive UI with **Lit** web components for lightweight, performant editor interface, and integrated **Winston** for structured logging of relay performance and session analytics.